Saturated fusion systems on groups of order 3^8

a running report

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Abstract

We search for the saturated fusion systems $\mathcal{F}$ on the groups S of order 3^8 with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$, in the manner of Parker–Semeraro [**?**] but with the fast proto-essential search of Gautam [**?**] and the fusion-graph-free saturation test of the GAP package `FusionSystems` that accompanies this report. There are 1 396 077 groups of order 3^8 (Vaughan-Lee's database, available in GAP through the package `sglppow`). Section **??** reduces them to a short list by necessary conditions, Section **??** identifies the surviving groups, Section **??** contains the look-up tables of subgroups of $\mathrm{GL}_n(3)$ with a strongly 3-embedded subgroup, and Section **??** records the fusion systems found group by group. This document is updated as the computation proceeds; the current state is summarised in Section **??**.

Contents

1 Status

[updated as the work proceeds]

- Stage 1a (cheap reduction of the 1 396 077 groups): *running*.
- Stage 1b (automorphism group tests on the survivors): not started.
- Look-up tables for $\mathrm{GL}_n(3)$: $n \leq 3$ in progress, $n = 4$ queued.
- Stage 2 (fusion systems group by group): not started.

2 Conventions and the strategy

Throughout $p = 3$, S is a group of order 3^8 and $\mathcal{F}$ is a saturated fusion system on S with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$. We write $\mathrm{Aut}_S(E) = N_S(E)/C_S(E)$ and $\mathrm{Out}_S(E) = \mathrm{Aut}_S(E)\mathrm{Inn}(E)/\mathrm{Inn}(E)$; for an S-centric subgroup E (that is $C_S(E) \leq E$) we have $\mathrm{Out}_S(E) \cong N_S(E)/E$.

By the Alperin–Goldschmidt fusion theorem $\mathcal{F}$ is generated by $\mathrm{Aut}_{\mathcal{F}}(S)$ and the groups $\mathrm{Aut}_{\mathcal{F}}(E)$ for the $\mathcal{F}$-essential subgroups E (the $\mathcal{F}$-centric, fully normalised subgroups with $\mathrm{Out}_{\mathcal{F}}(E)$ having a strongly 3-embedded subgroup). Following [**?**] we therefore proceed as follows.

1. Determine the subgroups $E \leq S$ which can be essential in some saturated fusion system on S (the *proto-essential* subgroups), up to $\mathrm{Aut}(S)$-conjugacy. Here we use Gautam's theorem (Theorem **??** below) instead of the subgroup lattice of S.

2. Determine the Borel subgroups, i.e. the $3'$-subgroups $K \leq \mathrm{Aut}(S)$ up to conjugacy, so that $\mathrm{Aut}_{\mathcal{F}}(S) = \mathrm{Inn}(S)K$ (recall that $\mathrm{Inn}(S)$ is a normal Sylow 3-subgroup of $\mathrm{Aut}_{\mathcal{F}}(S)$).

3. For each Borel subgroup and each proto-essential subgroup E determine the candidates for $\mathrm{Aut}_{\mathcal{F}}(E)$: the subgroups of $\mathrm{Aut}(E)$ containing $\mathrm{Inn}(E)$ with $\mathrm{Aut}_S(E)$ as a Sylow 3-subgroup, with Out having a strongly 3-embedded subgroup and compatible with the Borel subgroup.

4. For each choice of a set of essential subgroups (up to the action of $N_{\mathrm{Aut}(S)}(\mathrm{Aut}_{\mathcal{F}}(S))$) and automizers, build the fusion system, discard it unless $O^3(\mathcal{F}) = \mathcal{F}$, $O_3(\mathcal{F}) = 1$ and $\mathcal{F}$ is saturated, and finally remove isomorphic copies.

Steps 2–4 are only carried out for the groups surviving the reduction of Section **??**. All computations are done with the GAP package `FusionSystems`; the scripts for this report are in the directory `order38/scripts` of the repository.

Theorem 2.1 (Gautam [**?**, Theorem 3.6])**.** *Let $S = L_1 > L_2 > \cdots > L_m = 1$ be a central series of S with all factors of order p which refines the lower central series. Every essential subgroup of a saturated fusion system on S is S-conjugate to a subgroup $C_S(xA/A) = \{s \in S : [s, x] \in A\}$ with $A = L_i \leq [S, S]$ and xA an element of order p of S/A.*

3 The reduction

3.1 Necessary conditions

We use the following necessary conditions on S. Conditions (R1)–(R3) do not involve automorphism groups and are applied to all 1 396 077 groups (Stage 1a); conditions (R4)–(R6) need $\mathrm{Aut}(S)$ and the automorphism groups of the candidates and are applied to the survivors (Stage 1b).

Lemma 3.1 ([**?**, Lemma 7.1])**.** (R1) *If $|S : \mathrm{Z}(S)| \leq 9$ then S carries no saturated fusion system with $O_3(\mathcal{F}) = 1$.*

Lemma 3.2. (R2) *Let E be an essential subgroup of a saturated fusion system $\mathcal{F}$ on S and $T = \mathrm{Out}_S(E) \cong N_S(E)/E$. Then*

(a) E is S-centric and $E < N_S(E)$;

(b) T is a Sylow 3-subgroup of a group with a strongly 3-embedded subgroup, so T is cyclic, or elementary abelian of rank at least 2, or $T \cong 3^{1+2}_+$, or $T \cong \mathrm{Syl}_3(\mathrm{PSU}_3(9))$ of order 3^6;

(c) $|E/\Phi(E)| \geq |T|^2$;

(d) if $1 = D_0 < D_1 < \cdots < D_k = E$ is a chain of characteristic subgroups of E, then no element of $N_S(E) \setminus E$ centralises every factor D_i/D_{i-1}.

Proof. (a) and (b) are part of the definition and the classification of the groups with a strongly p-embedded subgroup (for $p = 3$ and T non-cyclic the possibilities for $O^{3'}(\mathrm{Out}_{\mathcal{F}}(E)/O_{3'}(\mathrm{Out}_{\mathcal{F}}(E)))$ are $\mathrm{PSL}_2(3^n)$, $\mathrm{PSU}_3(3^n)$, ${}^2G_2(3^n)$, $\mathrm{PSL}_3(4)$ and M_{11}; the Ree groups have Sylow 3-subgroups of order at least 3^9). (c) is [**?**, Lemma 2.?]: $\mathrm{Out}_{\mathcal{F}}(E)$ acts faithfully on $E/\Phi(E)$ because the kernel of $\mathrm{Aut}(E) \to \mathrm{Aut}(E/\Phi(E))$ is a normal 3-subgroup and $O_3(\mathrm{Out}_{\mathcal{F}}(E)) = 1$ as E is $\mathcal{F}$-radical. (d) The automorphisms of E centralising all factors of a chain of characteristic subgroups form a normal subgroup K of $\mathrm{Aut}(E)$, and K is a 3-group (a $3'$-automorphism centralising all factors of a normal series of a 3-group is trivial). Hence $K\mathrm{Inn}(E)/\mathrm{Inn}(E) \leq O_3(\mathrm{Out}(E))$ meets $\mathrm{Out}_{\mathcal{F}}(E)$ in a normal 3-subgroup, which is trivial, so $K \cap \mathrm{Aut}_S(E) \leq \mathrm{Inn}(E)$, which is (d) as E is S-centric. □

In Stage 1a a subgroup E is called a *candidate* if it is S-conjugate to a subgroup $C_S(xA/A)$ of Theorem **??** and satisfies (a)–(d) of Lemma **??**, where in (d) we use the chain $E > \Phi(E) > 1$ and then the covering pairs of the set of characteristic subgroups generated (by one round of intersections and products) by $\mathrm{Z}(E)$, $\mathrm{Z}_2(E)$, $\Phi(E)$, E', $\gamma_3(E)$, $\mho_1(E)$, $\Omega_1(\mathrm{Z}(E))$, $\mathrm{Z}(\Phi(E))$ and $C_E(E')$. (The stability group of the set of covering pairs is contained in the stability group of any maximal chain in the set, so the argument of (d) applies.)

Lemma 3.3. (R3) *If $W \neq 1$ is a characteristic subgroup of S which is contained in and characteristic in every candidate, then $W \leq O_3(\mathcal{F})$ for every saturated fusion system $\mathcal{F}$ on S.*

Proof. W is normalised by $\mathrm{Aut}_{\mathcal{F}}(S)$ and by $\mathrm{Aut}_{\mathcal{F}}(E)$ for every essential E (an essential subgroup is S-conjugate to a candidate E_0, and $W = W^s$ is characteristic in E_0^s). By Alperin–Goldschmidt W is normalised by every morphism of $\mathcal{F}$ defined on it, i.e. $W \trianglelefteq \mathcal{F}$. □

In Stage 1a "characteristic" is replaced by the sufficient condition of membership in the sets of visibly characteristic subgroups described above.

Lemma 3.4. *Let $\mathcal{E}$ be a set of representatives of the S-classes of the proto-essential subgroups of S (the candidates which in addition satisfy $\mathrm{Aut}_S(E) \cap O_3(\mathrm{Aut}(E)) = \mathrm{Inn}(E)$ and, when $\mathrm{Out}_S(E)$ is not cyclic, $\mathrm{Aut}(E)$ not soluble; both conditions are $\mathrm{Aut}(S)$-invariant and a candidate whose $\mathrm{Aut}(S)$-orbit leaves the set of candidates is discarded).*

(R4) If $\mathcal{E} = \emptyset$ then $O_3(\mathcal{F}) = S$ for every saturated $\mathcal{F}$ on S.

(R5) If a nontrivial characteristic subgroup of S is characteristic in every member of $\mathcal{E}$, then $O_3(\mathcal{F}) \neq 1$.

(R6) If $S \neq \langle [S, O^3(\mathrm{Aut}(S))], E : E \in \mathcal{E} \rangle$ then $O^3(\mathcal{F}) < \mathcal{F}$ for every saturated $\mathcal{F}$ on S.

Proof. (R4) and (R5) are as before. (R6): $O^3(\mathcal{F}) = \mathcal{F}$ if and only if the hyperfocal subgroup $\langle [P, O^3(\mathrm{Aut}_{\mathcal{F}}(P))] : P \leq S \rangle$ is S, and by Alperin–Goldschmidt it is generated by $[S, O^3(\mathrm{Aut}_{\mathcal{F}}(S))]$ and the $[E, O^3(\mathrm{Aut}_{\mathcal{F}}(E))] \leq E$ for E essential; finally $O^3(\mathrm{Aut}_{\mathcal{F}}(S)) \leq O^3(\mathrm{Aut}(S))$. □

3.2 Statistics

[to be filled in when Stage 1a finishes]

4 The surviving groups

[to be filled in]

4.1 Sylow 3-subgroups of simple groups of order 3^8

The finite simple groups G with $|G|_3 = 3^8$ and non-abelian Sylow 3-subgroups are (up to the obvious isomorphisms):

- $\mathrm{Alt}(18)$, $\mathrm{Alt}(19)$, $\mathrm{Alt}(20)$ (and the symmetric groups), with $S \cong (C_3 \wr C_3) \times (C_3 \wr C_3)$;
- $\mathrm{PSp}_4(9)$, S a Sylow 3-subgroup of $\mathrm{Sp}_4(9)$;
- $\mathrm{PSL}_3(q)$ with $3^4 \parallel q - 1$ (e.g. $q = 163$) and $\mathrm{PSU}_3(q)$ with $3^4 \parallel q + 1$ (e.g. $q = 647$), with $S \cong \big(\{(x, y, z) \in C_{81}^3 : xyz = 1\} \rtimes C_3\big)/C_3$ of maximal class;

- ${}^3D_4(q)$ with $3^3 \parallel q^2-1$ (e.g. $q=53$), where S is an extension of $C_{81}\times C_{27}$ by C_3 (not constructed here).

The groups $\mathrm{PSL}_2(3^8)$, $\mathrm{PSL}_2(q)$ with $3^8 \parallel q\pm 1$ and $\mathrm{PSp}_4(q)$ with $3^4 \parallel q^2-1$ have abelian Sylow 3-subgroups, which carry no fusion system with $O_3(\mathcal{F})=1$. No sporadic group has $|G|_3=3^8$. The first three groups are constructed in `scripts/known_groups.g`; their invariants are

S	$\lvert\mathrm{Z}(S)\rvert$	class	rank	exponent	$\lvert S'\rvert$	classes
$\mathrm{Syl}_3(\mathrm{Alt}(18))$	9	3	4	9	81	289
$\mathrm{Syl}_3(\mathrm{PSp}_4(9))$	9	3	4	9	81	161
$\mathrm{Syl}_3(\mathrm{PSL}_3(163))$	3	7	2	81	729	737

5 Look-up tables: subgroups of $\mathrm{GL}_n(3)$ with a strongly 3-embedded subgroup

If $E\cong 3^n$ is an elementary abelian essential subgroup then $\mathrm{Out}_{\mathcal{F}}(E)=\mathrm{Aut}_{\mathcal{F}}(E)\le \mathrm{GL}_n(3)$ has Sylow 3-subgroup $T=\mathrm{Aut}_S(E)$ of order at most 3^{8-n} (as E is centric and $N_S(E)>E$, in fact $n\le 7$) and a strongly 3-embedded subgroup. The following tables list, for each $\mathrm{GL}_n(3)$-class of admissible T (cyclic, elementary abelian or 3^{1+2}_+), the subgroups H with $T\in\mathrm{Syl}_3(H)$ and a strongly 3-embedded subgroup, up to conjugacy in $N_{\mathrm{GL}_n(3)}(T)$. For $n\le 4$ the tables are complete. They are produced by `scripts/gltables.g` and stored in machine-readable form in `data/gln_table.g`.

[tables to be included: `tables/gl2.tex`, `tables/gl3.tex`, `tables/gl4.tex`]

6 Results group by group

[to be filled in: one subsection per group, in the order of the computation (groups without large elementary abelian subgroups first), with a table of the fusion systems found in the style of [**?**]: for each $\mathcal{F}$ the group $\mathrm{Out}_{\mathcal{F}}(S)$, the essential subgroups with their automizers, and an identification or the remark "exotic?"]

References

[1] C. Parker, J. Semeraro, *Algorithms for fusion systems with applications to p-groups of small order*, Math. Comp. 90 (2021), arXiv:2003.01600.

[2] P. Gautam, *Fusion systems on Sylow 3-subgroups of Fischer and Monster sporadic groups II*, arXiv:2607.24674.

[3] C. Broto, N. Castellana, J. Grodal, R. Levi, B. Oliver, *Subgroup families controlling p-local finite groups*, Proc. LMS 91 (2005).